ArtBench-10 Generative Modeling: VAE, DCGAN, and Diffusion

Nuno Batista and Miguel Cabral Pinto

University of Coimbra

Abstract. We design, train, and evaluate multiple generative model families on ArtBench-10: a VAE, a DCGAN (and cGAN variant), and diffusion-based models, including fine-tuning of a pretrained Google DDPM. Models are selected via Bayesian hyperparameter sweeps on a 20% subset and evaluated with FID/KID. Final results are reported for full-dataset training with evaluation repeated across 10 random seeds.

1 Introduction

We study generative modeling on ArtBench-10, a 10-class dataset of 32×32 paintings [1]. We implement and compare a VAE [2], a DCGAN and conditional variant (cGAN) [3], and diffusion-based models [4], including fine-tuning a pretrained Google DDPM. We evaluate sample quality with FID and KID and report results across 10 random seeds.

2 Problem Description

The target domain is ArtBench-10 [1], a curated benchmark of 50,000 training and 10,000 test images across 10 artistic styles. Each image is a 32×32 RGB painting.

The objective is to learn a generative model $p_\theta(x)$ that can sample plausible paintings while maintaining fidelity and diversity. Key challenges include:

- **Fidelity:** Avoiding overly smooth or noisy samples.
- **Diversity:** Covering multiple styles without collapse.
- **Stability:** Handling adversarial (GAN) and iterative (diffusion) training dynamics.

3 Methodology and Approach

3.1 Data Pipeline

The data pipeline handles the ArtBench-10 dataset (32×32 RGB images, 10 styles). Images are converted from integer pixels to floating point and normalized to the range $[-1, 1]$ via

$$x' = 2 \cdot \frac{x}{255} - 1.$$

This normalization centers the data around zero and standardizes the dynamic range across models, which improves numerical conditioning for gradient-based optimization and aligns the data scale with common generative-model output parameterizations (e.g., a `tanh` output layer naturally produces values in $[-1, 1]$). For diffusion models, using a symmetric, zero-centered range is also convenient because the forward noising process is defined around zero-mean Gaussian noise.

During the experimentation phase, a 20% subset (10,000 images) was used for hyperparameter sweeps via Bayesian optimization. Once optimal parameters were identified, the final models were retrained on the full 50,000-image training set.

3.2 Model Families

The architectural design choices for each model family were tailored to the 32×32 resolution. Table 1 provides a comparative summary of the structural components across all implemented networks.

Table 1. Comprehensive architectural summary of the implemented generative models.

Feature	VAE	DCGAN / cGAN	Diffusion	Google DDPM
Backbone	4-Layer CNN	4-Layer CNN	2-Stage UNet	4-Stage UNet
Upsampling	Transpose Conv	Transpose Conv	Transpose Conv	Upsample + Conv
Downsampling	Strided Conv	Strided Conv	Strided Conv	Strided Conv
Normalization	Batch Norm	Batch Norm	Group Norm	Group Norm
Activation	LeakyReLU (0.2)	ReLU (G) / LReLU (D)	SiLU (Swish)	SiLU (Swish)
Bottleneck	$2 \times 2 \times 512$	$1 \times 1 \times 64$	$8 \times 8 \times 512$	$4 \times 4 \times 512$
Conditioning	N/A	Class Embedding	Time + Class	Timestep
Attention	None	None	Bottleneck (Opt)	Multi-head

VAE Our VAE implementation features a symmetric 4-layer convolutional architecture (32→16→8→4→2). Each layer employs Batch Normalization and LeakyReLU activations. We employ the reparameterization trick to allow for end-to-end backpropagation through the 256-dimensional stochastic bottleneck.

DCGAN and cGAN The GAN architectures follow DCGAN [3]. The cGAN variant incorporates a style embedding (dimension 64) concatenated with the noise vector in the generator and injected as an additional channel in the discriminator.

Diffusion (From Scratch) Our "TinyUNet" consists of two downsampling and two upsampling stages using Residual Blocks with Group Normalization and SiLU activations. Skip connections preserve high-frequency stylistic details.

Diffusion (Pretrained Google DDPM) The `google/ddpm-cifar10-32` model [5] was fine-tuned, which uses a larger UNet with self-attention layers.

3.3 Implementation and Reproducibility

Our pipeline is configuration-driven via YAML files. We utilize Weights & Biases (W&B) for real-time metric logging. All experiments were conducted using PyTorch, with random seeds fixed to 67 for hyperparameter comparisons.

3.4 Theoretical Analysis of Hyperparameters

We summarize the swept hyperparameters and their primary effects on optimization stability, capacity, and sample quality.

Variational Autoencoders (VAE)

- **Learning rate (`lr`):** Controls optimization step size; too large can destabilize training and degrade reconstructions, while too small slows convergence.
- **Latent dimension (`latent_dim`):** Sets bottleneck capacity; larger values increase expressiveness but can weaken regularization and encourage blurry reconstructions if the decoder dominates.
- **Base channels (`base_channels`):** Scales network width and capacity; increasing it typically improves fidelity at higher compute cost.
- **KL weight (β):** Scales the KL regularizer in the VAE objective, i.e., it weights the pressure for $q_\phi(z \mid x)$ to match the prior $p(z)$. With a reconstruction loss term (here, effectively an $L2$-style distortion term) and a KL term, increasing β reduces the information capacity of the latent (it discourages encoding fine detail in z), typically improving sampling from the prior and latent smoothness but worsening reconstruction fidelity. Decreasing β does the opposite: it behaves more like a deterministic autoencoder ($\beta \to 0$), often improving reconstructions but producing a latent distribution that deviates from the prior, which can harm unconditional sample quality when sampling $z \sim p(z)$.
- **Batch size (`batch_size`):** Affects gradient noise and effective learning rate; larger batches stabilize optimization but reduce stochastic regularization.

Generative Adversarial Networks (GAN)

- **Learning rate (`lr`):** One of the most sensitive GAN hyperparameters; higher values can cause divergence or oscillations in the discriminator and generator game.
- **Latent dimension (`latent_dim`):** Controls the dimensionality of the generator input noise; too small can limit diversity, while too large can make mapping harder without improving quality.
- **Base channels (`base_channels`):** Changes generator/discriminator capacity; imbalance can lead to either an overpowered discriminator (vanishing gradients) or an underpowered discriminator (poor signal).

- **Discriminator updates (`d_updates_per_g`):** Sets the ratio of discriminator steps per generator step; increasing it can stabilize training when the discriminator is weak, but can also over-train the discriminator and stall generator learning.
- **Spectral normalization (`use_spectral_norm`):** Constrains discriminator Lipschitzness, often improving stability, but can reduce discriminator flexibility and slow learning depending on the regime.
- **Observed regime:** The best setting we found used `d_updates_per_g`=1, `use_spectral_norm`=false, and `latent_dim`=64. Samples remained prone to high-frequency artifacts consistent with transpose-convolution upsampling.

Conditional GAN (cGAN)

- **Generator conditioning (concat in latent space):** Given a noise vector z (dimension d) and a class label y, the generator embeds the label with an embedding layer $e(y)$ (dimension 64) and concatenates them $[z; e(y)]$. The concatenated vector is reshaped to $(d + 64) \times 1 \times 1$ and fed through the standard DCGAN transposed-convolution stack. This makes the class signal available from the first layer onward.
- **Discriminator conditioning (label as an image channel):** The discriminator embeds y into a vector of size $32 \cdot 32$, reshapes it into a $1 \times 32 \times 32$ map, and concatenates it with the RGB image as an extra input channel (input becomes $4 \times 32 \times 32$). The discriminator therefore learns to score joint pairs (x, y) rather than marginal realism of x.
- **Effect on optimization:** Conditioning increases discriminator capacity (it can use the label map as a shortcut feature). If the discriminator learns class-specific cues faster than the generator can match them, gradients to the generator become less informative, reducing both diversity and conditional fidelity.

Diffusion Models

- **Learning rate (`lr`):** Controls how quickly the denoiser fits the noise-prediction objective; too large can harm stability or lead to overshooting, while too small slows convergence.
- **Base channels (`base_channels`):** Sets UNet width; larger models typically improve sample fidelity and style capture at increased compute.
- **Attention (`use_attention`):** When enabled, the TinyUNet inserts a multi-head self-attention block at the bottleneck feature map (after the mid residual block, at the lowest spatial resolution). Concretely, it computes Q, K, V via a 1×1 convolution, flattens the $H \times W$ spatial grid into tokens, and applies scaled dot-product attention so each location can aggregate information

from all other locations. The attention output is projected back with a 1×1 convolution and added residually. This can improve global coherence (e.g., consistent structure across distant regions) at relatively low cost because the bottleneck has the smallest $H \times W$.

- **Training length:** For the from-scratch model, extending training to 50 epochs produced the largest quality gains.
- **Fine-tuning:** For Google DDPM, a higher learning rate (5×10^{-4}) improved adaptation from CIFAR-10 priors to ArtBench.

4 Experimental Setup

4.1 Hyperparameter Search

Hyperparameter selection was performed on a **20% training subset** (10,000 images) using **Bayesian optimization** (Weights & Biases sweeps). Each sweep run trained a candidate configuration and evaluated sample quality using **FID** (minimize). For computationally expensive diffusion-based models, sweep runs used reduced evaluation settings (e.g., fewer generated samples and shorter training) and the best configuration was then retrained with the final evaluation settings. Our stopping criteria, though varying across models, was around the point where the best FID plateaued for a significant number of consecutive runs.

We used model-specific search spaces and budgets, as seen in Table 2:

Table 2. Hyperparameter search spaces for each model's Bayesian sweep

Model	Parameter	Values
VAE	`lr`	{0.00005, 0.0001, 0.0005, 0.001, 0.002}
	`latent_dim`	{64, 128, 256}
	`base_channels`	{64, 128}
	β	{0.5, 1, 2, 4}
	`batch_size`	{64, 128, 256}
DCGAN / cGAN	`lr`	{0.00005, 0.0002, 0.001}
	`latent_dim`	{64, 100, 128}
	`base_channels`	{64, 128}
	`d_updates_per_g`	{1, 2, 5}
	`use_spectral_norm`	{true, false}
Diffusion	`lr`	{0.00005, 0.0002, 0.0005}
	`base_channels`	{64, 128}
	`use_attention`	{true, false}
Google DDPM	`lr`	{0.00001, 0.00005, 0.0002, 0.0005, 0.01}
	`epochs`	{10, 20, 40}

To visualize the explored hyperparameter configurations and their performance trends, we used Weights & Biases sweep diagnostics. Figure 1 shows **parallel coordinates** plots (one per model family). Additionally, we performed **fANOVA-based importance analysis** to estimate which parameters most strongly influenced FID; Figure 2 displays the importance summaries (one per model family).

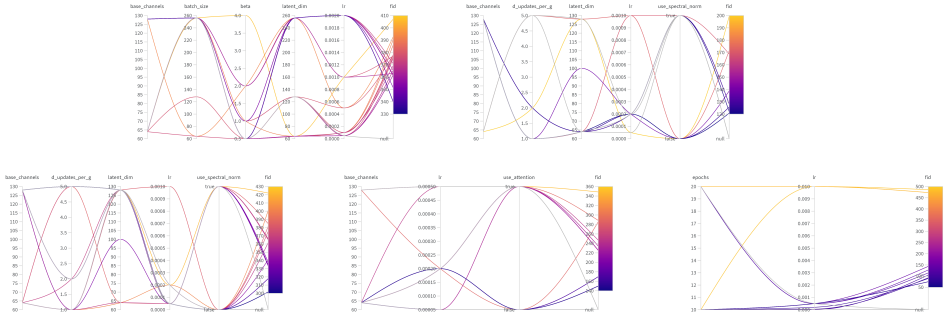

Fig. 1. Weights & Biases parallel coordinates plots for the hyperparameter sweeps.

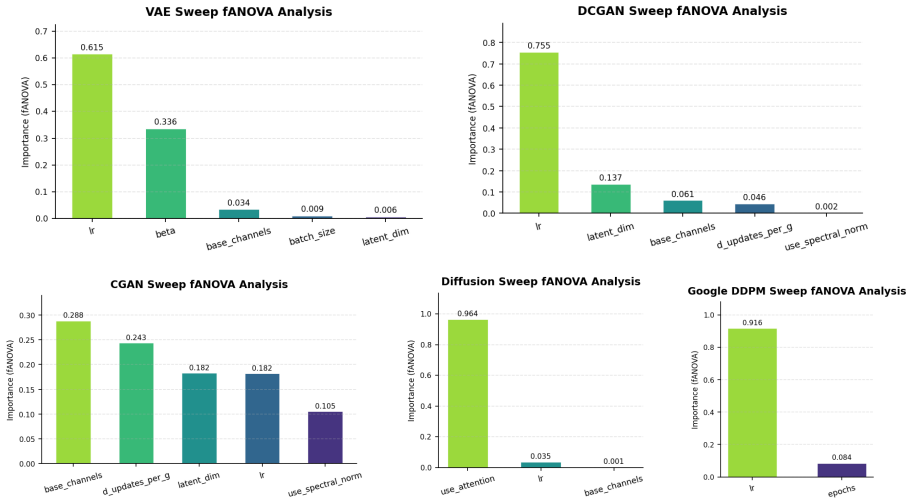

Fig. 2. fANOVA parameter importance for the hyperparameter sweeps, reporting the estimated contribution of each hyperparameter to FID variance for each model family.

Several patterns emerge from the sweeps. Learning rate is the dominant factor for most models, with the parallel plots confirming that high learning rates uniformly degrade performance. An exception is the **Diffusion model**, where the binary `use_attention` flag alone explains 96.4% of variance, and the parallel coordinates plot shows a clean split, with all low-FID runs passing through

`use_attention=false`. At 32×32 resolution, the additional complexity of self-attention appears to impair rather than help optimization. For the **VAE**, the KL weight `beta` is the second most important parameter, which is expected given its direct control over the reconstruction–regularization trade-off; the sweep selected a low value ($\beta = 0.5$), prioritizing reconstruction fidelity. The **DCGAN** sweep found `use_spectral_norm` nearly irrelevant (0.2%), suggesting that at this scale the standard adversarial objective is already stable enough. In contrast, the **cGAN** importance profile is remarkably flat across all five parameters (range 0.105-0.288), indicating no single knob governs training outcome, a sign that the model's failure (seen in the results section) is systemic rather than attributable to any specific misconfiguration. For the **Google DDPM**, the parallel plot reveals a clear trend: lower learning rates and more training epochs consistently yield lower FID, with `lr=0.01` leading to catastrophic divergence (FID ≈ 500).

Table 3 shows the configuration with the lowest FID obtained for each model.

Table 3. Final hyperparameter configurations for full-dataset training.

Parameter	**VAE**	**DCGAN**	**CGAN**	**Diffusion**	**Google DDPM**
Learning Rate	0.002	0.0002	0.00005	0.0002	0.0005
Batch Size	256	128	128	32	32
Latent Dim.	256	64	128	128	128
Base Channels	128	128	128	128	128
Optimizer	Adam	Adam	Adam	AdamW	AdamW
Beta (β)	0.5	-	-	-	-
D. Updates per G.	-	1	5	-	-
Spectral Norm	-	No	No	-	-
Attention	-	-	-	No	-

4.2 Experimental Protocol

After each sweep, we used the hyperparameters in Table 3 to train the models on the 20% subset. We then performed a **final evaluation** by training that best configuration once (seed 67) and computing FID/KID using **5,000 generated** samples versus **5,000 real** samples (KID computed over 50 random subsets of size 100), with the evaluation repeated across **10 different random seeds** for statistical robustness (we report Mean ± Standard Deviation). Finally, we retrained all model families on the **full 50k** training set using their best hyperparameters and repeated the same 10-seed evaluation.

5 Results and Analysis

5.1 Qualitative Results

Visual inspection of 64-sample grids in Figures 3 and 4 reveals distinct model characteristics across the different families.

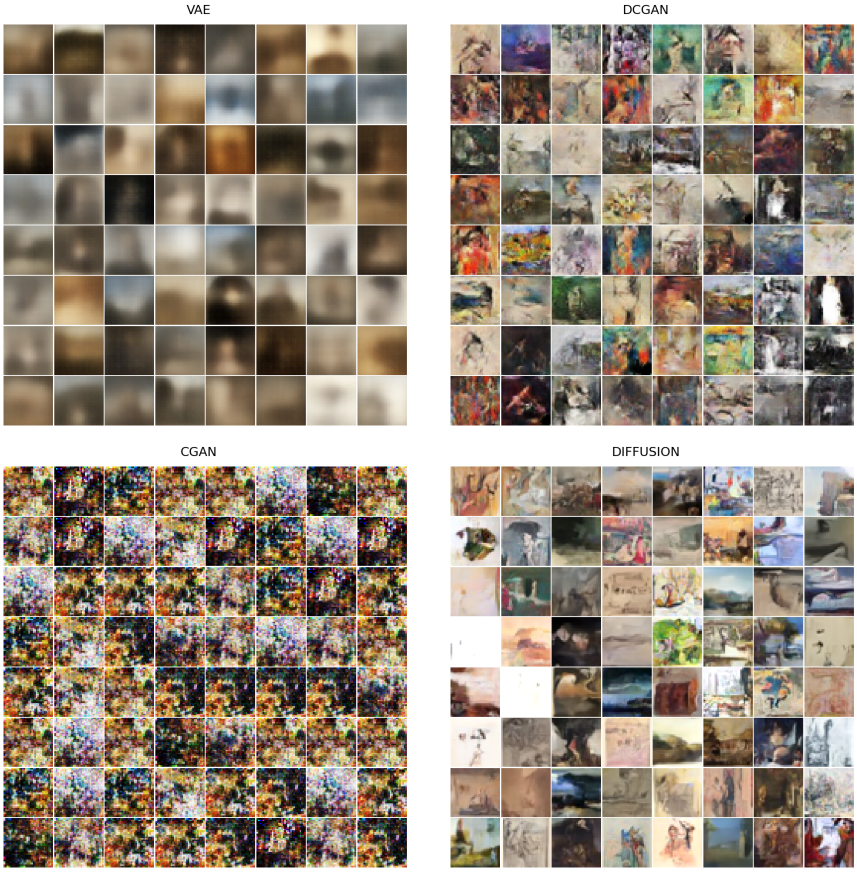

Fig. 3. Qualitative samples from the main model families evaluated on ArtBench-10.

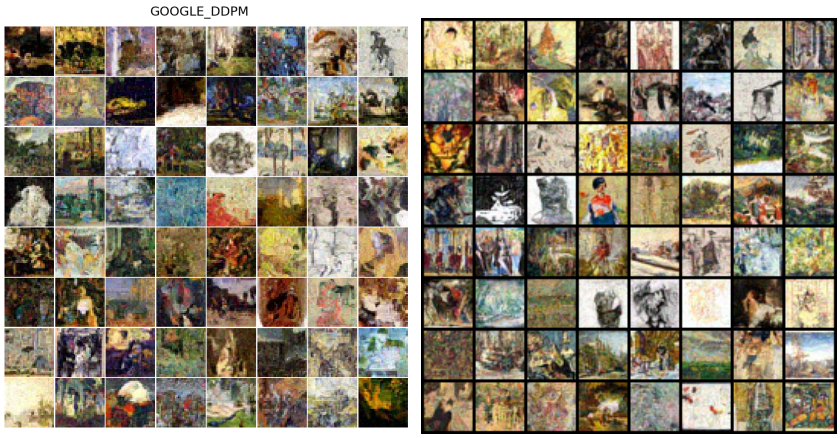

Fig. 4. Samples from the fine-tuned Google DDPM (left: 20 epochs / right: 200 epochs), demonstrating more recognizable shapes compared to the other models

5.2 Quantitative Results

Table 4 presents the results of the 10-seed evaluation on both the full dataset and the 20% subset used for hyperparameter selection.

Table 4. Quantitative results (Mean ± Std Dev) across 10 random seeds for the 20% subset and the full dataset (100%).

	20% Subset		**Full Dataset**	
Model	**FID** ↓	**KID** ($\times 10^{-3}$) ↓	**FID** ↓	**KID** ($\times 10^{-3}$) ↓
VAE	347.52 ± 0.73	336.1 ± 1.4	355.26 ± 1.32	370.3 ± 4.0
DCGAN	$\mathbf{121.87 \pm 0.70}$	$\mathbf{90.9 \pm 1.2}$	91.02 ± 0.65	69.0 ± 1.5
cGAN	301.88 ± 0.81	258.4 ± 1.7	393.68 ± 1.36	329.1 ± 2.6
Diffusion	176.58 ± 1.64	120.7 ± 2.8	$\mathbf{43.03 \pm 0.51}$	$\mathbf{26.7 \pm 1.0}$
Google DDPM	122.82 ± 1.30	121.3 ± 2.5	97.49 ± 1.26	90.5 ± 1.5

On the full dataset, FID and KID rankings are in perfect agreement across all five models, and per-seed standard deviations are small relative to mean values (e.g., FID std of 0.51 for TinyUNet, 0.65 for DCGAN), confirming the stability of both the models and the evaluation protocol. The large gap between TinyUNet (FID = 43.03) and the next best model DCGAN (FID = 91.02) underscores the advantage of diffusion-based objectives on this dataset.

On the 20% subset, the ranking shifts notably: DCGAN achieves the best KID ($90.9 \pm 1.2 \times 10^{-3}$) and the best FID (121.87), with Google DDPM close behind (FID = 122.82). Diffusion (TinyUNet) ranks third by FID (176.58) despite dominating on the full dataset, suggesting that data volume is a stronger determinant of diffusion model quality than for GANs. The relative ordering of VAE and cGAN remains poor in both settings, though cGAN shows a notable relative improvement on the subset (FID = 301.88 vs. 393.68 on the full dataset), which indicates less severe mode collapse when the per-class data distribution is smaller.

5.3 Training Dynamics

Loss curves measured over training across 10 seeds for the full dataset(Figure 5) highlight different optimization behavior across model families:

- **DCGAN:** Loss drops sharply in the first few epochs and stabilizes, indicating a relatively balanced adversarial equilibrium consistent with its competitive FID.
- **cGAN:** Loss increases monotonically throughout training (∼5 to ∼7.5) with a widening confidence band across seeds, indicating progressive divergence. This directly correlates with cGAN's poor FID of 393.68.

- **VAE:** Exhibits a transient spike near epoch 7-8 (rising to ∼0.4 before recovering to ∼0.1), with a wide per-seed confidence band indicating the instability occurs at different epochs across runs. Training ultimately stabilizes.
- **Diffusion (TinyUNet):** Noise-prediction MSE decreases smoothly and reproducibly, with a near-zero confidence band, reflecting a supervised regression-style objective [4]. This stability correlates with the best observed FID.
- **Google DDPM:** Starts from an already low loss (∼0.033) due to pretraining on CIFAR-10, and fine-tunes gradually. Despite the lower absolute loss values, it achieves a worse FID (97.49) than the from-scratch TinyUNet, suggesting domain mismatch rather than optimization failure.

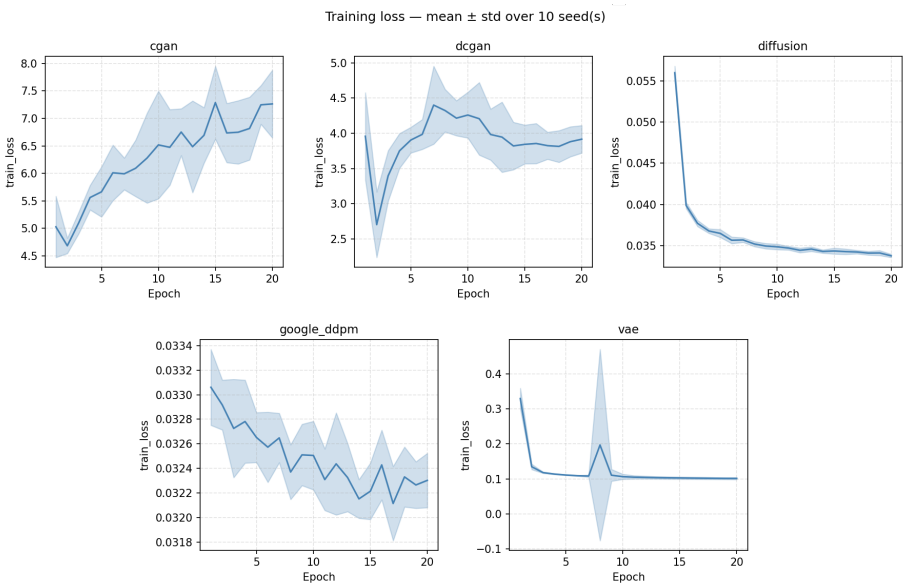

Fig. 5. Training loss curves (mean ± std over 10 training seeds) for all five model families.

5.4 Ablations and Discussion

Our results highlight several key trade-offs:

- **Unconditional vs. Conditional:** DCGAN outperformed cGAN significantly, which exhibits per-class mode collapse: the generator maps each class to a single fixed output, producing zero intra-class diversity. The collapsed outputs are themselves incoherent high-frequency noise, indicating the generator learned a degenerate per-class "solution" to fool the discriminator

without capturing any meaningful structure. This is consistent with the monotonically diverging loss. The class-conditioning mechanism appears to destabilize the adversarial objective under the chosen capacity and training budget.

- **Diffusion Dominance:** On the full dataset, diffusion models achieved the best FID by a large margin, benefiting from a stable MSE loss landscape and skip connections. On the 20% subset, however, it lost to other models, indicating they require more data to realize their generative advantage. Visually, they exhibit a wide diversity of styles, representing landscapes, figures, line drawings, and abstract pieces with convincing low-level texture throughout.
- **Perceptual Quality:** While TinyUNet Diffusion achieved the lowest global FID, the fine-tuned Google DDPM produced the most coherent samples by visual inspection despite a higher FID. Additionally, despite good FID and producing samples with visible texture and good color diversity, DCGAN samples suffer from checkerboard noise, worsening its perceived quality. These findings highlight that metrics do not always align with human perception of quality.
- **VAE smoothness:** VAEs optimized with an $L2$-style reconstruction loss optimize for the mean: when multiple high-frequency outcomes are plausible, the decoder is rewarded for predicting an average, which tends to wash out texture and contrast. This effect belongs to the model/loss combination (not a single sweep knob), and helps explain the extremely smooth samples, that loosely capture the global brownish color palette of the dataset but entirely lack fine detail.

Why Google DDPM can look better but score worse. FID and KID evaluate distributional similarity in a fixed Inception feature space rather than directly measuring perceived coherence [6]. The fine-tuned Google DDPM starts from a CIFAR-10 prior that encodes photographic structure; after fine-tuning on ArtBench, it generates structurally coherent images, but their visual character remains closer to natural photographs than to paintings. This stylistic misalignment is precisely what FID/KID penalize: the generated distribution diverges from the ArtBench painting statistics in texture and style cues, while human observers reward the structural plausibility. In addition, a model may produce high-quality samples while under-covering modes (reduced diversity), which is penalized by distribution-level metrics even when qualitative grids appear strong.

6 Conclusion

On ArtBench-10, diffusion-based models provided the best overall performance under our evaluation protocol. Quantitatively, the custom TinyUNet diffusion model achieved the lowest FID, while qualitatively the fine-tuned Google DDPM produced the most coherent samples.

Use of External Resources and AI Tools

This project utilized ArtBench-10 (Kaggle). We used the `diffusers` library and `google/ddpm-cifar10-32` (Hugging Face). Gemini assisted in LaTeX formatting and analysis verification.

References

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